

StoryDrawer

A Co-Creative Agent Supporting Children's Storytelling through Collaborative Drawing

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ABSTRACT

Storytelling is a common creative activity for children. During storytelling, children need creative support and are subjected to cognitive challenges. This paper explores a co-creative agent - StoryDrawer, which supports children in creating oral stories through collaborative drawing. StoryDrawer works with children in two strategies: Child says and AI draws; and Child scribbles and AI completes. These two collaborative strategies allow children to draw their stories as an externalization and provoke unexpected ideas. This paper presents the interaction design, collaborative strategies and implementation of StoryDrawer, and conducts a user study for the fun and task effectiveness of StoryDrawer. The results reveal that our system can encourage children's active participation in storytelling and help them create novel stories through human-computer collaboration.

CCS CONCEPTS

• Human-centered computing; • Collaborative and social computing; • Collaborative and social computing devices;

KEYWORDS

Storytelling, Oral expression, Co-creative system, Human-AI collaboration, Creativity support tool, Children

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1 INTRODUCTION

Interactive storytelling support has become an important area of concern for HCI [3]. Despite considerable efforts invested in supporting storytelling activities and increasing children's engagement in storytelling [21], much prior work ignores the creative support that children need and the cognitive burdens to which children are subjected during storytelling. To better support creative thinking and to alleviate children's cognitive load including fear of the blank canvas, writer's block, perfectionism and so on [13], we hypothesize that the incorporation of human-computer collaboration in storytelling would be beneficial because co-creative systems have been widely explored to effectively provide inspiration, provoke divergent thinking and support innovation [14].

With artificial intelligence (AI) development, co-creation systems have emerged, which bring AI and human collaborators together to engage in creative activities. However, few studies [12] focus on AI agents collaborating with children's storytelling and the procedures for building such systems. For instance, Sun et al. [22] proposed a mixture strategy and envisioned a robot that works with children to create stories by inserting new story content. In this paper, we explore how AI can collaborate with children and provide new opportunities to develop and share stories. We incorporate drawing in this process during collaborative storytelling because a shared drawing is considered a significant representation medium to externalize the mental images of children and encourage creative thinking [9]. Therefore, our work exploits the previous definition of co-creative systems, but with a collaborative drawing to support children's storytelling.

In this paper, we present StoryDrawer (Figure 1), an intelligent agent, embedded with an AI-based co-creation approach that supports children aged 5-10 to create oral stories. StoryDrawer includes a web-based digital storytelling application and a tangible button named IDEA. When interacting with StoryDrawer, users tell stories verbally, and the web application transforms the oral description



Figure 1: StoryDrawer System: (a) Child starts using StoryDrawer. (b) Child presses the IDEA button. (c) Child shows his/her work.

into drawings. We encourage children to scribble during the storytelling process. StoryDrawer will inspire children if they get ‘stuck’ by generating new sketches based on what they have drawn after the children press the IDEA button. To evaluate the fun and task effectiveness of StoryDrawer, we conducted a user study using both quantitative and qualitative methods. The results show that children reap the fun of interacting with StoryDrawer, and our system can help children create more novel stories. The main contributions of our work are:

- We presented a novel intelligent agent, integrated with the co-creative system, to support children’s storytelling by transforming their oral stories into sketches and generating new doodles based on what the children have drawn.
- We proposed two collaborative strategies: Child says, and AI draws, and Child scribbles, and AI completes. These help children build their inner world’s external structure and inspire children to create oral stories.
- We conducted a user study to examine the task effectiveness of StoryDrawer and to measure children’s fun.

2 RELATED WORK

In this section, we review previous studies to understand co-creativity and the potential of child-AI interaction in supporting children’s storytelling.

2.1 Storytelling

Prior work has confirmed that storytelling can effectively help children improve their language, logical thinking, communication and imagination [27]. Due to the importance of this, numerous storytelling supporting tools have been presented. For example, KidPad [7] is a collaborative storytelling tool enabling children to create stories by drawing, typing and hyperlinking in a large two-dimensional zoomable space. Moreover, Wayang Authoring [24] is a web-based platform assisting children in creating stories using digital puppets. These existing storytelling tools primarily enrich the storytelling experience of children [8, 23]. In this paper, we contribute to the approach aimed at inspiring children, making it easier for them to create and illustrate stories by enabling AI to collaborate with children in storytelling.

2.2 Co-creativity

In designing creative systems and AI agents, more researchers have focused on co-creativity [1]. In domains like music [19], drawing

[4, 26] and education [11, 29], co-creativity has been studied. Co-creative systems have been widely applied to effectively provide inspiration, provoke divergent thinking and support innovation [14]. However, previous studies have focused on how to support the creativity of adults. There is also great potential in the field of child-AI collaboration. Based on these findings, we hypothesize that co-creativity can be applied in a specific domain, such as storytelling, to inspire children and help them create more novel stories.

2.3 Child-AI interaction

AI has begun to affect the way children live, learn and play [2, 16]. For example, the new generation of AI chatbot dolls (AIBD) is a mobile application with which children interact by text-messaging [17]. Williams et al. [25] developed a platform, PopBots, to improve children’s AI literacy. Children can build their own LEGO robot characters and can click on the tablet to interact with PopBots. Nicoletta et al. [18] presented an activity to disseminate the basic principles of coding in a fun and accessible way. In terms of storytelling, WAWLT [12] is a mixed-initiative, co-creative storytelling game in which two players develop a story transcript by selecting and editing actions. In this paper, we combine the results of these prior works and further explore the interaction between AI and children. We propose two interactive approaches, Child says, and AI draws, and Child scribbles, and AI completes, to support children in creating oral stories.

3 STORYDRAWER

3.1 Concept

Storytelling stimulates children’s imagination and visualization, which are the key to comprehension and higher-order thinking [5]. We define StoryDrawer as a creative and collaborative system. While children are speaking, StoryDrawer generates a real-time illustration of the story on the screen. During this process, the children take a leading role. In addition, we have designed two collaborative strategies, which can inspire children and help them externalize and reflect on their stories.

3.2 Interaction

Combining the concept of human-AI collaborative generation, we focus on integrating the StoryDrawer system seamlessly for children’s storytelling. Figure 2 shows an example workflow of StoryDrawer.

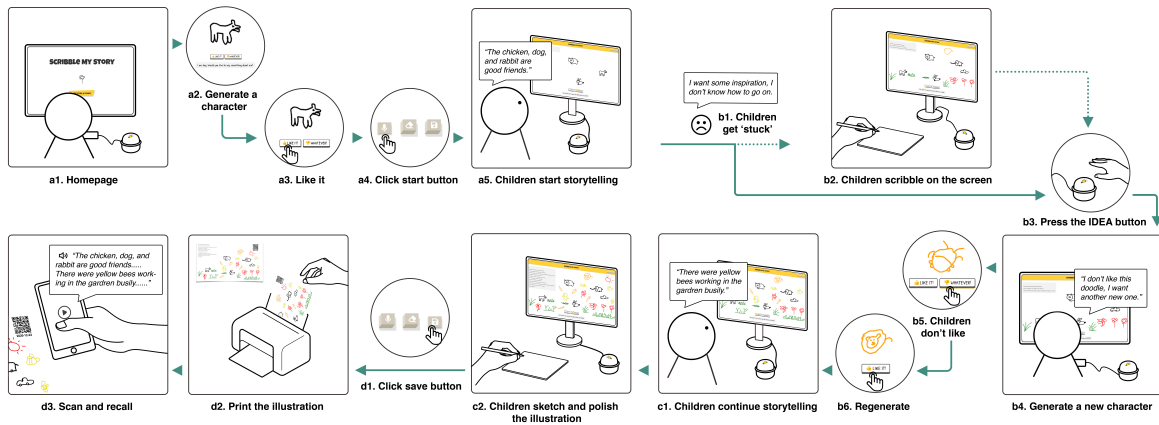


Figure 2: An example workflow of StoryDrawer: (a1-5) Let's tell a story. (b1-6) Getting 'stuck'. (c1-2) I want to polish it. (d1-3) It's a fantastic piece.

3.2.1 Let's Tell a Story. The child clicks 'LET'S TELL A STORY' on the homepage to enter the agent (Figure 2a1), then a random story element doodle will appear on the screen (Figure 2a2). We encourage the child to start with this element. This approach can help the child overcome the fear of having a blank canvas or blank pages in the creative process [13]. If the child does not like the initial element we provide, the child can click on the "WHATEVER" button to get a new one. If the child chooses "LIKE IT!" then the child can click on the record button and start telling a story (Figure 2a3-4). During storytelling, the agent analyzes the child's oral description and extracts keywords to generate the story illustration in real-time (Figure 2a5).

3.2.2 Getting 'Stuck'. When the child gets 'stuck' during storytelling (Figure 2b1), he/she can conveniently use an iPad (a line of tablet computers designed, developed and marketed by Apple Inc) to scribble on the screen (Figure 2b2). After scribbling, the child can press the IDEA button for inspiration (Figure 2b3). StoryDrawer will generate a sketch based on the child's drawings (Figure 2b4). Additionally, the child can also choose to press the IDEA button immediately to obtain a new story element doodle. In relation to the provided 'idea', the child gives feedback by selecting 'LIKE IT' or 'WHATEVER' (Figure 2b5). Furthermore, the 'WHATEVER' means that the agent will provide another 'idea' and will repeat the selection process (Figure 2b6).

3.2.3 I Want to Polish It. In addition to oral expression, the child can draw anything to refine and complement the story's illustrations (Figure 2c1-2). We give the child this opportunity to perfect the drawings and to make the whole image more compatible with the storytelling world in mind.

3.2.4 It's a Fantastic Piece. After finishing storytelling, the child clicks the save button and saves his/her work (Figure 2d1). The masterpiece includes story text on the top left, a story scene in the middle and a QR code on the top right. The child can print and share it with parents or friends (Figure 2d2). By scanning the QR code, parents can review the storytelling and the drawing process with their children (Figure 2d3). According to prior studies, combining

a voice review and playback in storytelling would help children develop more complex stories [5].

3.3 Collaborative Strategies

3.3.1 Child Says and AI Draws. In the process of storytelling, children need to create a series of entities, mixing and composing them in an improvised way [15]. Building and interacting with these entities' external structures can ease cognitive challenges in creative activities [10]. These externalizations can be doodles, diagrams, gestures or other external representations [9]. We choose doodles because this form can express the inner world of a child as an instinctive expression of behavior [28]. Therefore, StoryDrawer would recognize the child's speech and would generate the corresponding number and color of the doodles on the screen in real-time based on the child's oral description. For example, when the child expresses 'I have two red flowers', the agent will extract 'two', 'red', 'flowers' and will draw two red flowers on the screen. The real-time, voice-driven drawing will serve as an externalization and will help children project their inner story world into the real world.

3.3.2 Child Scribbles and AI Completes. Even when the creative process is going well, it is inevitable for children to suddenly find themselves at an impasse, unable to think of what step they should take to continue storytelling. This condition is often referred to as "writer's block" [13], and when it occurs, it can be disconcerting for children. In this case, we help the child think outside the box by providing the child with new story elements. When children are 'stuck', it is feasible for them to scribble on the screen. After the child presses the IDEA button, StoryDrawer predicts what the child wants to draw and generates figurative sketches, based on the child's scribbles, providing inspiration for the story in a collaborative drawing style.

3.4 Implementation

StoryDrawer is composed of a voice-driven drawing module and an inspirational generation module.

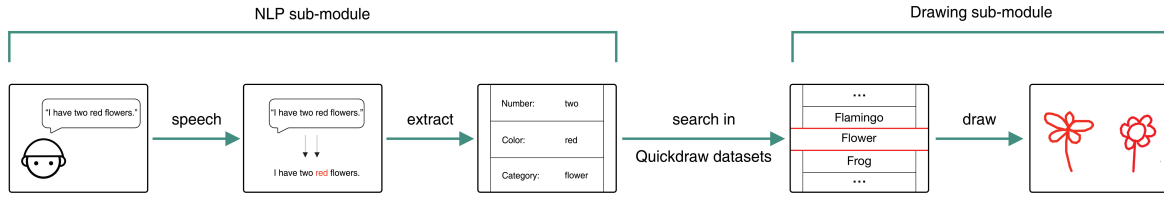


Figure 3: Workflow of voice drawing method.

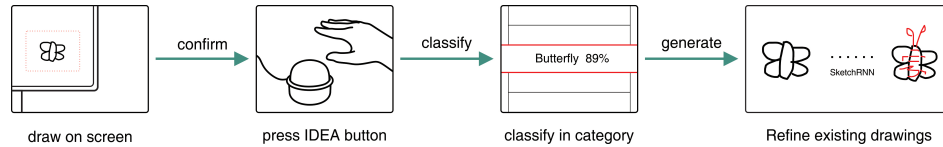


Figure 4: Workflow of idea generation method.

3.4.1 Voice-Driven Drawing Module. The voice-driven drawing module, which is composed of the natural language processing (NLP) sub-module and drawing sub-module, can analyze the child’s oral description and draw the story’s illustration in real-time (Figure 3) when storytelling. The NLP module transforms each sentence to corresponding text with Python when the children describe their stories verbally. Should a child speak a language other than English, the NLP module would translate it into English. Then, the NLP module extracts the nouns, amounts and colors from the text and sends them to the drawing sub-module. Finally, the drawing sub-module will map the nouns in the Quick Draw Dataset [4] and will draw doodles of corresponding quantity and color.

3.4.2 Inspirational Generation Module. The creativity generation module interacts with children through the IDEA button. The child can either press the IDEA button after scribbling or press it immediately for inspiration. After each press, a story element will appear randomly on the screen to provide the child with new inspiration. The IDEA button connects to the computer via a USB port. We adjust the Sketch-RNN model [4] and use P5.js to implement our interaction mode: 1) detect the stroke list drawn by the user on the screen; 2) identify target objects drawn by the child; 3) input the stroke list into the RNN model and generate new vector graphics. This algorithm can create more than 100 different types of sketches, based on input. Figure 4 shows the whole process of the inspirational generation module.

4 USER STUDY

To examine the task effectiveness of StoryDrawer and measure fun with children, we conducted a user study. We explored children’s experience, the resulting stories, and how the children collaborated with this agent in storytelling.

4.1 Participants and Procedure

We recruited 10 children (5F, 5M, mean age = 7.8, SD = 1.32) from a neighborhood school. They had never used any co-creation tools but had accessed story creation activities from the school curriculum. Additionally, three teachers with at least five years’ experience of working in children’s education, were recruited to score the stories

created in our study. The whole process was recorded with the consent of the children’s parents.

The study had a within-subject design in which participants created an oral story with StoryDrawer (the experimental condition) and without StoryDrawer (the control condition), respectively. The experiment was conducted in a classroom. Firstly, we described the functions of StoryDrawer and made sure they knew how to use the iPad. Then we invited each child to complete two oral story creation tasks. Each creation session (including storytelling and drawing) lasted 10 minutes, with a 20-minute interval between them. To prevent carryover effects, the orders of conditions were counterbalanced. During this session, two observers recorded the participants’ behavior. After the creation sessions, we conducted a short semi-structured interview with the children regarding their overall experience and story work. Then we utilized the Smileyometer and the Again-Again table, published by Read et al. [20] to measure the level of fun experienced by the children.

Each final story was rated in terms of integrity and novelty by the three experienced teachers. We presented all the final output stories in random order, and the teachers were asked to rate these stories in two dimensions, according to their experience using visual analog scales (VAS). The minimum score was 0 and the maximum score was 10, representing the lowest and the highest mark. Story integrity relates to whether the story has a complete story structure, while novelty is determined by the teachers’ subjective definition of novelty when assessing each story [6].

4.2 Quantitative Results

We confirmed fun with children in relation to StoryDrawer through the Smileyometer strategy and the Again-Again table. The results showed that only one of ten children considered our agent not very good, while the other nine children made positive comments with regard to StoryDrawer (Figure 5a). We used the Again-Again table to determine whether children would like to use StoryDrawer again. Eight children indicated that they would like to use StoryDrawer again, while two children were unable to decide at this time (Figure 5b). In general, StoryDrawer can encourage children to enjoy the process of storytelling.

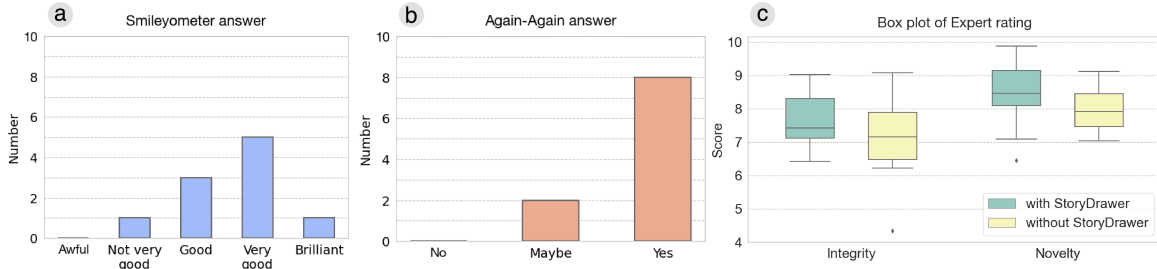


Figure 5: (a) Bar plot of Smileyometer answers. (b) Bar plot of Again-Again answers. (c) Box plot with the data distribution and the result of the paired sample t-test of an expert rating in integrity and novelty.



Figure 6: (a) Interview with users. (b) Examples of child-AI collaborative drawings in the user study.

To interpret the teachers' ratings of the integrity and novelty aspects of children's stories, we illustrate the data distribution by boxplot in Figure 5c. As the lines of medium values indicate, children's stories created with StoryDrawer scored higher from the perspective of integrity and novelty than stories not created with StoryDrawer. A Shapiro-Wilk test showed that the expert ratings were distributed normally. We then conducted a correlated samples t-test to analyse the data. The results showed that in terms of novelty, the scores in the experimental condition (mean score=8.5030) were significantly higher than in the control condition (mean score=7.9983, $t=2.449$, $p<.05$). However, in terms of integrity, there was no statistically significant difference.

4.3 Qualitative Results

We conducted a short semi-structured interview with the participants (Figure 6a). Figure 6b shows some of the collaborative drawings created by the participants, using StoryDrawer.

When the children were asked what role StoryDrawer played in the creation of their stories, C1 said, "It tells stories with me, like my friend." C4 said, "It's like teacher Yang. She is my favorite teacher. She taught us to draw our own stories." However, C8 saw it as more of a tool and said, "It helps me draw as I tell the story." C3 said, "It's like a magic book." Generally, younger children considered StoryDrawer a partner in their tasks, while older children treated the agent more as a tool.

A somewhat surprising finding was that one participant may be button-dependent. C7 pressed the IDEA button repeatedly to get a new doodle. This agent may also have an enlightening effect on children's learning of sketching. C1 expressed the desire to take

the IDEA button home and to use it to practice sketching. Most of the feedback on the IDEA button was positive, but C6 said: "I don't like it very much, it. Many of the doodles it painted are useless."

During our observation, most children, namely, C2 and C4, enjoyed the speaking process more. C2 did not experience the process of painting at all and dictated his own story throughout. C5 was more inclined to enjoy the painting process or was more interested in identifying and utilizing the painting process. This phenomenon shows the different interests of children when using StoryDrawer.

In general, we found that children expressed a keen interest in StoryDrawer, which played an active role in the storytelling process.

5 LIMITATIONS AND FUTURE WORK

We identified certain limitations in our research. Firstly, the story screen's overall layout lacks a sense of story, and the current agent only supports single-player creation mode. Secondly, the experiment's duration is not long enough for the system to develop with the children's feedback and thus lack the observation of long-term companionship and cooperation. We also lack user research relating to the psychological, cognitive process of children's collaboration with AI.

These limitations also suggest several avenues for future technical work. Firstly, we will enable this agent to make semantic connections and to generate connected story prompts, based on what the children have previously said. Furthermore, we wish to explore and design a multi-person, co-creation model. Additional research should examine the extent to which children rely on the IDEA button. Finally, we intend to evaluate the user experience

and conclude design principles for designing child-AI, co-creation systems.

6 CONCLUSIONS

In this paper, we propose StoryDrawer, an intelligent AI agent, which supports children's storytelling by collaborative drawing. It can analyze the child's oral description in real-time and extract essential information from it. Based on this information, StoryDrawer transforms children's stories into sketches and projects their imagination into the real world. In addition, we provide an IDEA button which children use to create drawings of story elements, based on their abstract doodles when they are 'stuck'. Our user study shows that StoryDrawer plays an active role in collaborative storytelling and that children can tell more innovative stories. Compared to previous storytelling supporting tools, we take inspiration from the concept of co-creativity and expand the realm of child-AI collaboration.

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